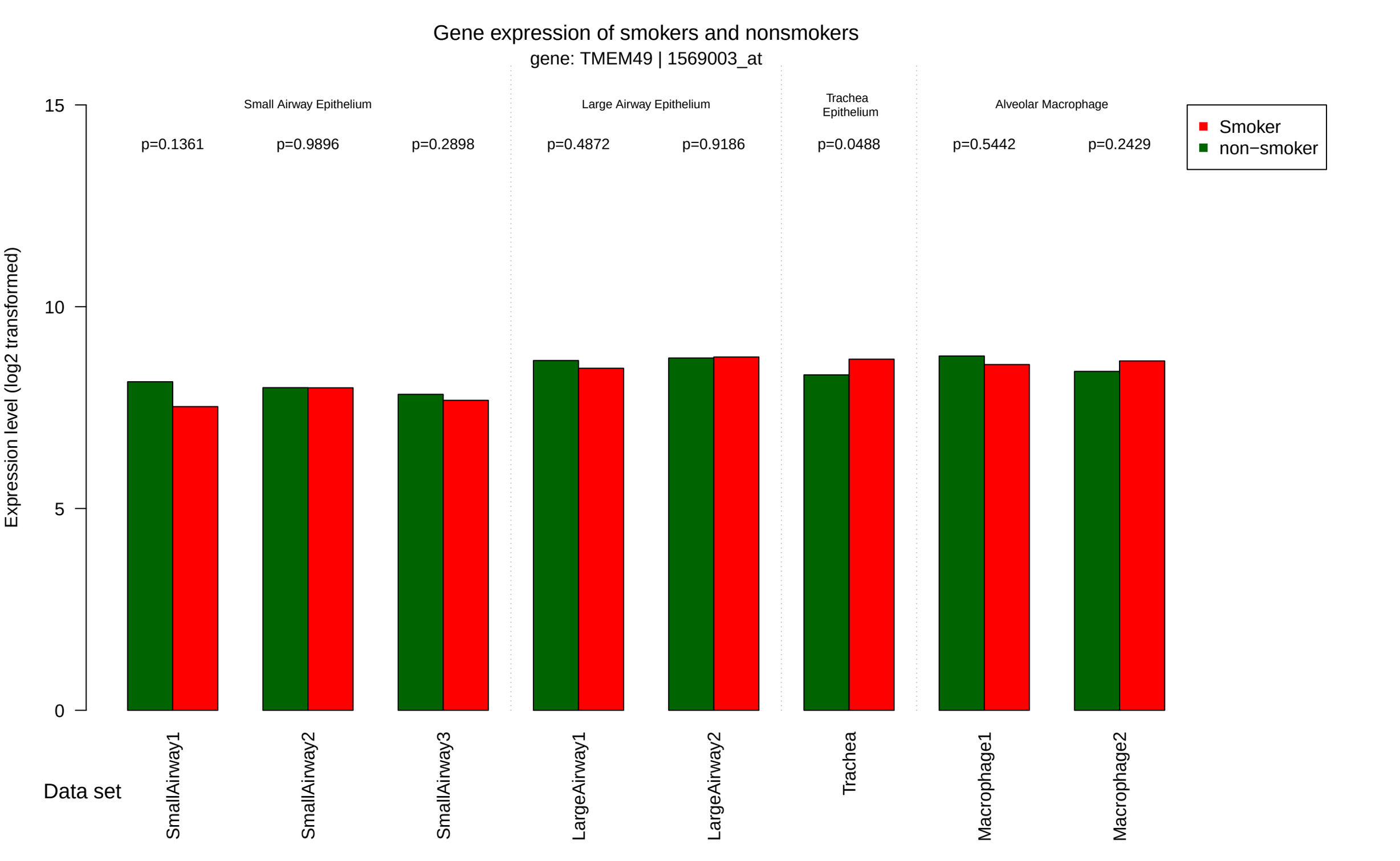
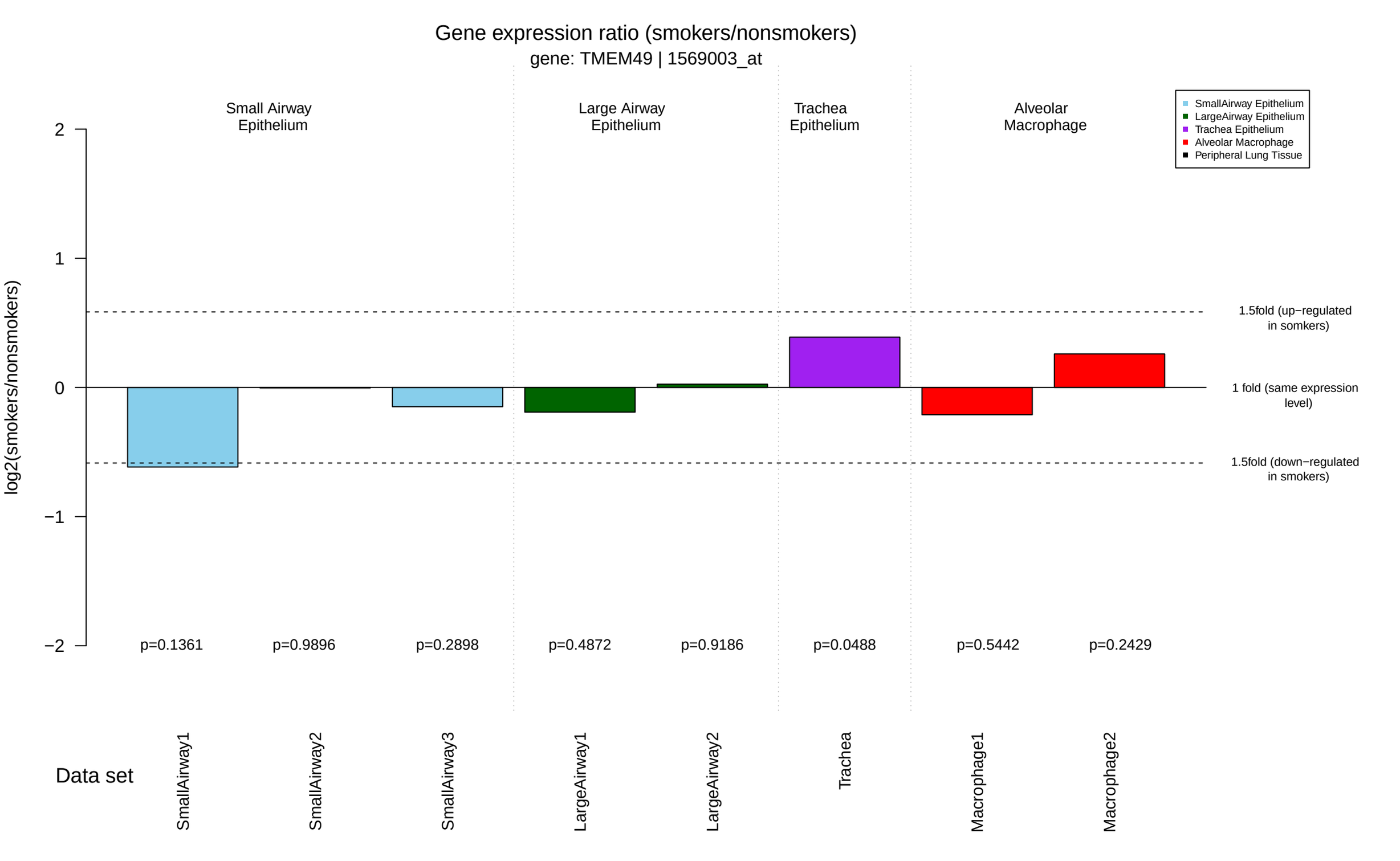
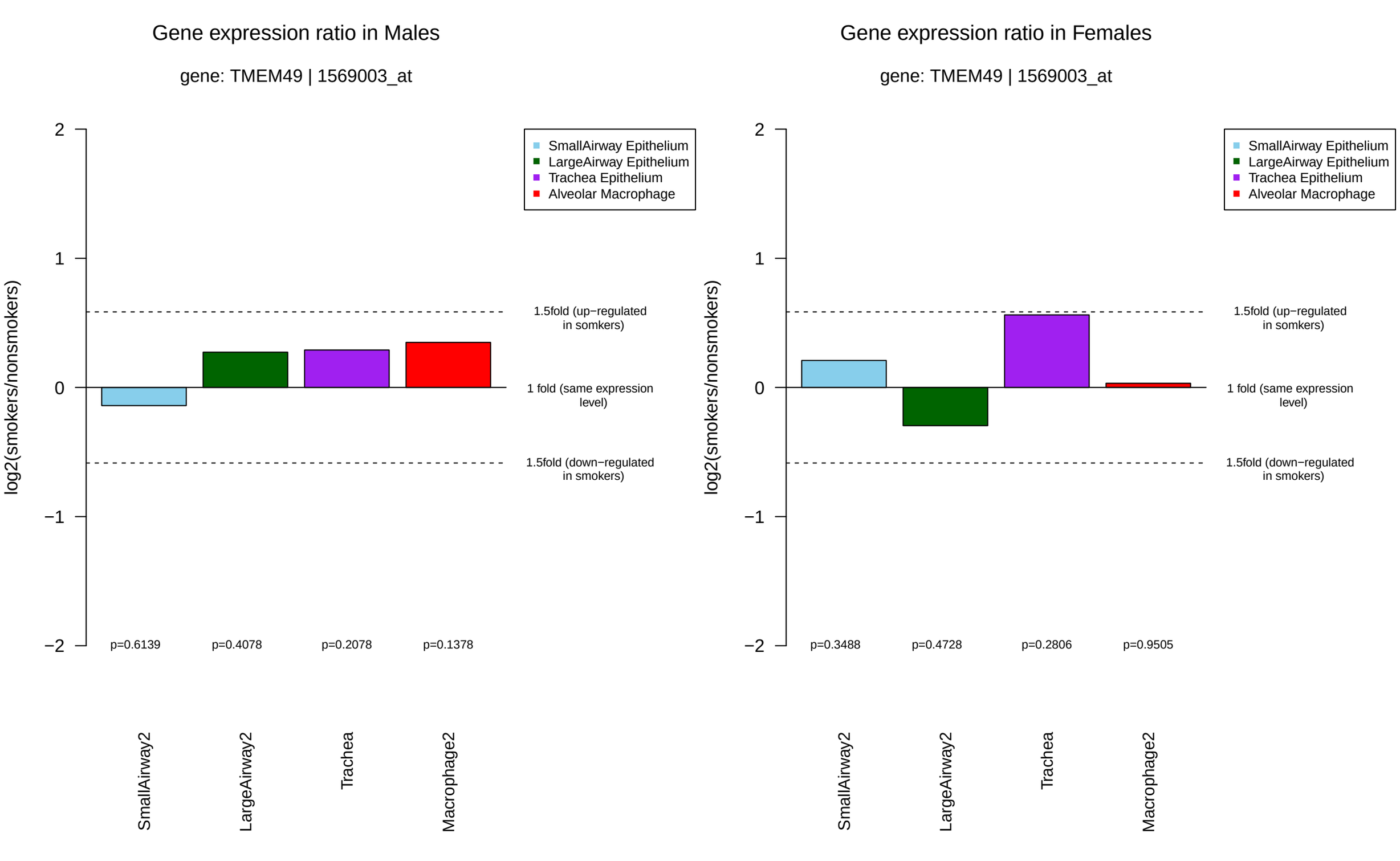
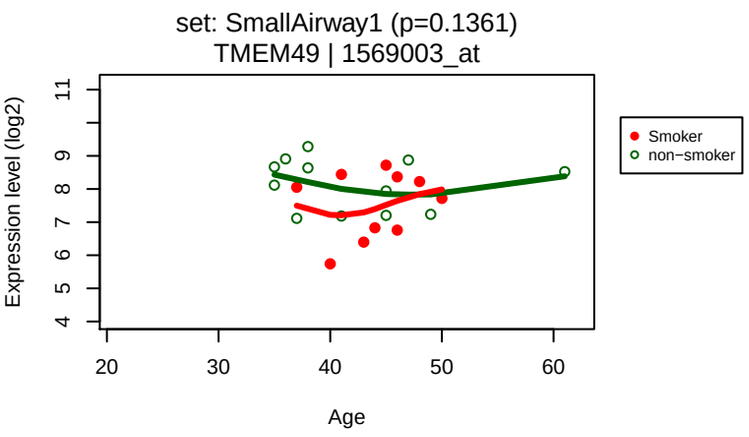
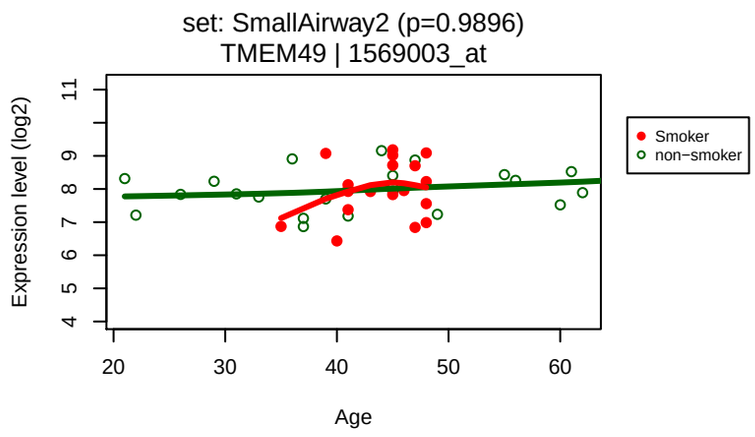
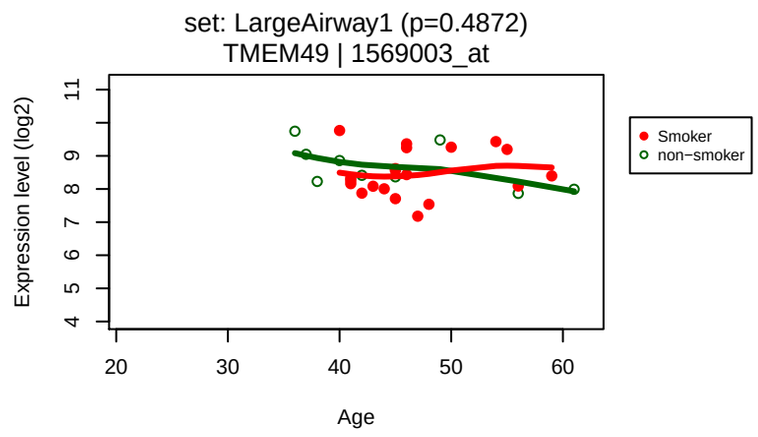

Gene expression of smokers and nonsmokers
gene: TMEM49 | 1569003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1361
p=0.9896
p=0.2898
p=0.4872
p=0.9186
p=0.0488
p=0.5442
p=0.2429
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

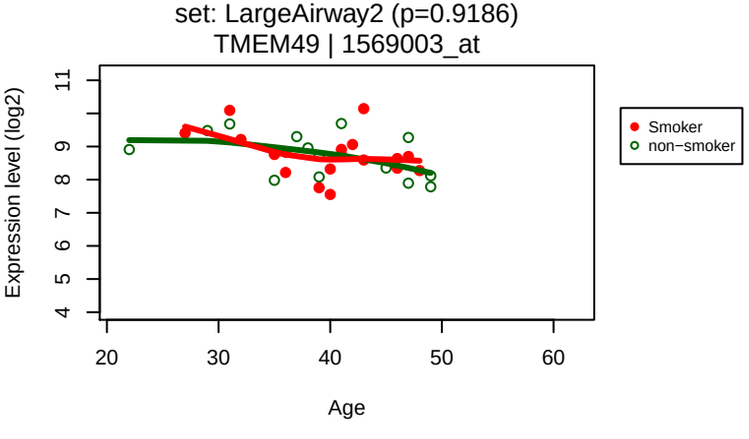

Gene expression ratio (smokers/nonsmokers)
gene: TMEM49 | 1569003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1361
p=0.9896
p=0.2898
p=0.4872
p=0.9186
p=0.0488
p=0.5442
p=0.2429
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

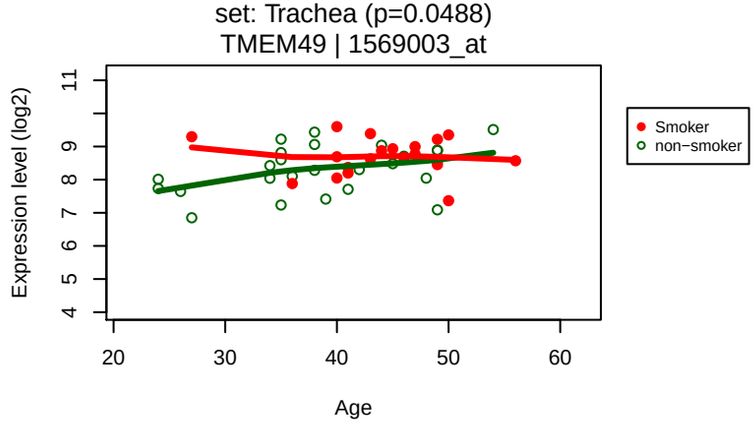

Gene expression ratio in Males
gene: TMEM49 | 1569003_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6139
p=0.4078
p=0.2078
p=0.1378
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM49 | 1569003_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3488
p=0.4728
p=0.2806
p=0.9505
SmallAirway2
LargeAirway2
Trachea
Macrophage2

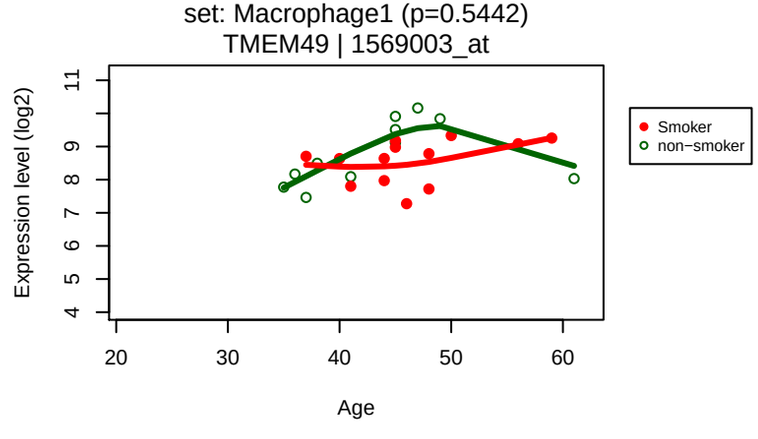

set: SmallAirway1 (p=0.1361)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker

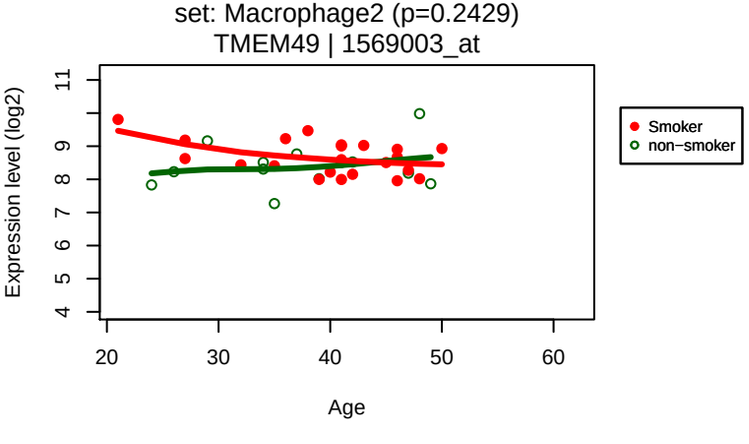

set: SmallAirway2 (p=0.9896)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker

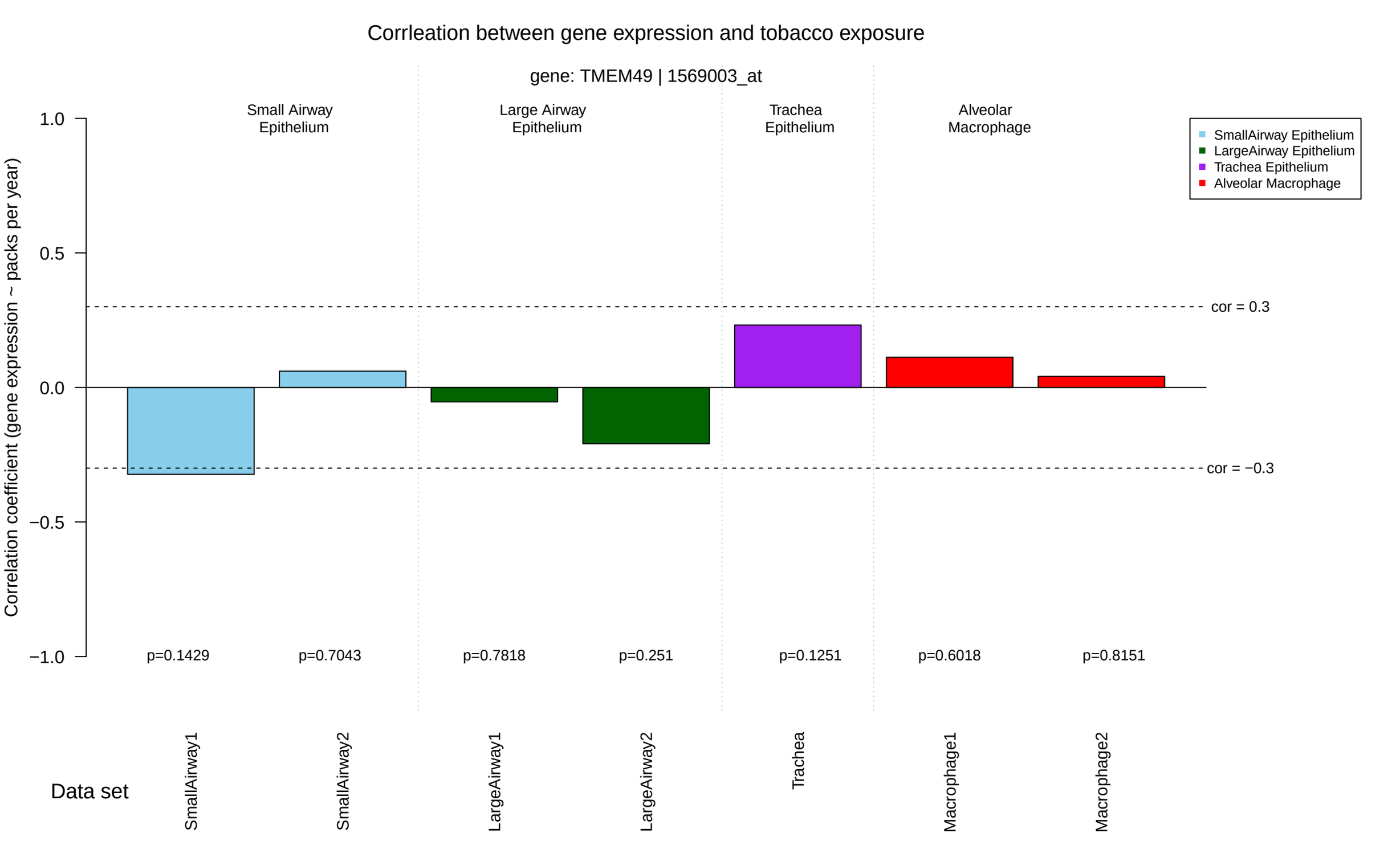

set: LargeAirway1 (p=0.4872)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9186)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0488)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5442)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2429)
TMEM49 | 1569003_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM49 | 1569003_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1429
p=0.7043
p=0.7818
p=0.251
p=0.1251
p=0.6018
p=0.8151
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2